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Preliminary Design and Performance Assessment of Screw-propelled Vehicle

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Abstract

Our star system is composed of different planets and moons, ranging from sun-scorched Mercury to Titan, where lakes and rivers, filled with liquefied methane, exist. Such huge differences in environments impose huge variation on the design of rovers overall and the mobility system in particular. In the current timeline, wheeled solutions are being used widely. Unfortunately, such solutions usually impose strict requirements on surface material properties, require suspension systems, complex control systems, and offer little in terms of flexibility and applicability to different environmental conditions. In this article, the use of a screw-propelled system is proposed. The system interfaces with the medium through one or several rotating auger-like cylinders with helical flanges, which engage with the medium on which the vehicle is moving and provide propelling force. The cylinder is attached directly to the vehicle's frame through bearings on each end, and each cylinder can rotate independently from the others. Such systems have already been demonstrated in Earth conditions, where they are used as vehicles for traversing difficult terrain and offer capabilities exceeding even tracked vehicles. Particular operational examples of such vehicles include, but are not limited to: ZIL-26091, a Soviet vehicle for recovery of re-entered Soyuz capsules, and Ice Challenger, which have crossed Bearing Strait in 2002. The advantages of the envisioned system: (I) simplicity of the system; (II) enhanced power and redundancy; (III) enhanced stability and low ground pressure; (IV) robustness of the system; (V) enhanced maneuverability; (VI) all-terrain capabilities. The study is focused on performing preliminary design and performance assessment of an exploration vehicle's mobility system for the Titan environment because its diverse terrain, consisting of methane lakes, mountains, dunes, and plains, enables demonstration of versatility of the chosen propulsion method. Standard engineering approaches, including performance estimation by reduced-order models, usage of specific masses, mass fractions, and specific powers based on already existing systems to derive mass-power budget, are being used.

Abbreviations

DEM	Discrete Element Method
EDL	Entry, Descent and Landing
HGSL	Helical Granular Scaling Laws
MMRTG	Multi-Mission Radioisotope Thermoelectric Generator
MSA	Marsh Screw Amphibian
RFT	Resistive Force Theory
RUC	Riverine Utility Craft
VCi	Vehicle Cone Index
WES	U.S. Army Engineer Waterways Experiment Station

1 Introduction

The technology of screw propulsion, also known as Archimedean screw propulsion, has a long history on Earth. The first patent for a screw-propelled vehicle was issued to William Lyttleton in 1794 [1], and the first operational screw-propelled vehicle, a twin screw-propelled steamboat ferry, was built by Colonel John Stevens and operated on the Hudson River in New York State in 1804 [1].

After that, a number of screw-propelled vehicles have been constructed in different countries over the years. In the USA, during the 1960s - 1970s, the Army conducted a thorough test program called Marsh Screw Amphibian (MSA) to obtain a new mobility system for

the marsh environment [2]. The trials were conducted on the Chrysler vehicle built, and the M29C "Weasel" tracked vehicle was used as an opponent [3].

During the 1960s, the soviet truck manufacturer Zil had developed and built the Zil 2906, a screw-propelled vehicle designed for the search and rescue of cosmonauts. Then, throughout the 1970s, the new version, Zil 29061, was made. That version is still used today by the Russian space agency Roscosmos.

To better understand the advantages and disadvantages against other types of mobility, it is necessary to describe the principles on which screw-propelled vehicles operate. In Fig. 1, a schematic representation of the forces acting on the rotors for four different directions of motion is shown for the vehicle with two screws.

On soft soils, the helical flanges are pushed into the soil by the vehicle's mass. When the rotors rotate, the helical flanges push against the soil and produce lateral (F_{lx} and F_{rx} in Fig. 1) and longitudinal (F_{lz} and F_{rz} on Fig. 1) forces. Lateral forces are canceled by means of rotating helices in opposite directions (e.g., the right helix rotates clockwise while the left helix rotates counter-clockwise), while longitudinal forces are added up (upper row on Fig. 1). If we made all helices rotate in one direction, our vehicle would begin to turn around its center of mass (lower row on Fig. 1).

On hard surfaces, when the helical flanges are not

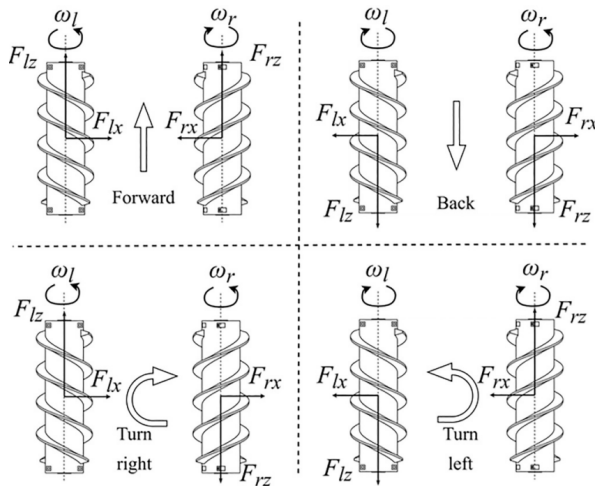


Fig. 1: Screw-drive propulsion direction. [4]

pushed in the medium, they start to act like regular wheels, causing the vehicle to move in a lateral direction [5]. Screw-propelled vehicles have several advantages compared to other types of mobility [5]. First of all, they can traverse difficult terrains such as soft, muddy ground or deep snow. Moreover, the rotors on which the helical flanges are placed add buoyancy, while the helical flanges create thrust, giving the vehicle the ability to float and move in a liquid. Due to the nature of the interaction between the helical flange and the soil, a significant tractive force could be developed. Another advantage is the simplicity of the mobility system compared to other variants built for difficult terrain, for example, tracked vehicles.

An Army test report on MSA [3] states that a screw-propelled vehicle is superior to a tracked vehicle (M29C) on extremely soft soils.

However, disadvantages are also present [5]. Screw-propelled vehicles have low maximum speed and require high energy due to significant losses from friction between the soil and the surfaces of the rotor and flange. Traversing the screw-propelled vehicle over a hard surface could cause damage to the surface or flanges. In addition, the Army test report on MSA [3] has noted the inability to execute severe maneuvers (e.g., sharp turns) and to be towed.

Due to the signal delays introduced by the distance between Titan and Earth, rovers designed for Titan shall have a high level of autonomy, which by itself limits the maximum speed of the craft. So, it could be said that the disadvantages of low maximum speed and inability to execute sharp turns are not so pronounced for such robotic missions.

Terrestrial experience with screw-propelled vehicles, terramechanics formulations, and mission studies for Titan are individually available in the literature, but they have not yet been combined into a method suitable for preliminary design. In particular, screw propulsion has not been assessed within a consistent sizing loop for a planetary mission, and, as a result, it cannot yet be com-

pared with wheeled or tracked solutions in Phase 0/A trade-offs in terms of mass, power, and feasibility margins.

The present study addresses the following research question: how can a preliminary design method quantify the mass, power, and mobility performance of a screw-propelled rover for Titan-like terrain while accounting for the coupling between vehicle sizing and screw-terrain interaction? To this end, the paper provides (i) an iterative mass-closure loop that includes a reduced-order screw-terrain model, (ii) a vehicle database in which reported values are distinguished from assumptions, (iii) a comparison of the screw-terrain model and of the sizing loop with the full-scale trafficability tests of the MSA, from which a screw traction efficiency is derived, and (iv) a Titan case study, including a sensitivity analysis and a feasibility box. The study does not aim to demonstrate that screw propulsion outperforms wheeled or tracked mobility on Titan; rather, it provides a means of addressing this question for a given mission.

The paper is organized as follows. Section 2 reviews related work, and Section 3 discusses use cases. Section 4 presents the methodology, the Titan case study, and the validation. Section 5 discusses the results, and Section 6 summarizes the conclusions.

2 Related Works/Studies

Terrestrial screw-propelled vehicles. The first analytical treatment of screw propulsion is attributed to Cole, who studied single-start helices mounted on hollow drums in a soil tank in 1961; this work remains the basis of most subsequent models [2]. The US Army program of the 1960s led to MSA, which was tested by WES on sand, silt, and fat clay and compared with M29C Weasel [3, 6, 7]. The tests established a 50-pass vehicle cone index (VCI) of 5 on fine-grained soil with free water, compared to approximately 25 for the tracked vehicle. They also reported a towing force of about 24% of the test weight on sand and 11% on wet clay, immobilisation on clay with a dry surface, and limited obstacle-crossing capability on the dikes [3]. Subsequent model tests at the Davidson Laboratory showed that the ratio between drawbar pull and weight increases as load decreases, since the soil failure surface develops below the helix rather than at the ground surface [2]. The larger Riverine Utility Craft (RUC), the Soviet ZIL-2906 and ZIL-29061 recovery vehicles, the Japanese Garinko amphibious vehicles, and the more recent HELIX robots extend the same architecture to heavier vehicles and to snow and ice [2]. However, none of these programs published a screw-terrain model together with the corresponding vehicle data, and Villacrés et al. conclude that the available dynamic models of screw propulsion remain subject to substantial limitations [2].

Screw-terrain modelling. Bekker-derived models of screw propulsion adapt the pressure-sinkage and shear relationships of wheel terramechanics to a helix. According to Villacrés et al., they remain applicable only

while the sinkage is less than the flight height, that is, while the screw operates on top of the soil rather than within it [2]. Thoesen et al. measured the thrust generated by submerged double-helix screws in glass beads and compared it with discrete element method (DEM) simulations, obtaining discrepancies of 5 to 20%, while resistive force theory (RFT) did not reproduce the distribution of force along the screw [2, 8]. The same group subsequently proposed helical granular scaling laws (HGSL), which predict the velocity and power of a scaled screw within 3 to 9% on the BP-1 lunar simulant [9]. Chen et al. characterised screw locomotion in water and in dry, wet, and saturated sand, and found that blade height dominates performance in granular media, pitch dominates in water, and screw stalls in saturated sand [10]. Sagara et al. measured the sideslip of screw wheels on a 20° slope of silica sand [11]. These studies provide parameter sensitivities and validity limits, but each of them characterises a single screw rather than a complete vehicle.

Planetary screw-propelled concepts and Titan mobility. CASPER is a 3.4 kg four-screw rover that excavates approximately 30 kg of regolith simulant per hour with a power consumption below 30 W [12]. ARCSnake combines Archimedean screw modules in a reconfigurable snake-like robot and reports a module mass of 1.0 kg for a 128 mm by 196 mm segment including its drive, which is the only measured screw module mass identified in the present review [13]. For Titan, Genta and Genta sized a 40 kg wheeled rover on lunar-regolith parameters, with a SNAP-19 power source, a speed of 100 m/h, a 20° grade, and a 100 mm obstacle [14]. The Titan Aerial Daughtercraft study provides atmospheric and battery assumptions [15], and the TANDEM tensegriety rover study provides the 213 kg lander and 12 kg payload benchmark adopted here [16]. The mechanical properties of the Titan surface are known only at the Huygens landing site, by penetrometry [17].

Research gap. To the authors' knowledge, no published work combines terrestrial evidence, screw-terrain models, and the Titan mission context in a form that allows a screw-propelled option to be included in a Phase 0 trade-off alongside wheeled and tracked vehicles. The present paper addresses this gap by means of a closure loop that includes the screw-terrain block, a database that separates reported values from assumptions, a validation against the only full-scale test campaign that reports geometry, weight, and performance together, and a Titan case study with the associated sensitivity analysis.

3 Use Cases

Our current knowledge of Titan is limited to data mainly obtained by Cassini-Huygens. The scientists have compiled a list of scientific goals that have to

be accomplished to enhance our understanding of Titan. These goals include studying the atmosphere of Titan, geology, lakes and rivers, and assessing habitability [18]. For that purpose, several systems have been envisioned, ranging from stationary landers to balloons and drone swarms.

The screw-propelled craft could be used as a stand-alone exploration vehicle or in conjunction with balloons to perform direct measurements on the surface. Its all-terrain capabilities would be particularly useful for exploring the shore side of Titan's lakes, where the medium's properties are between regolith and liquid methane. The ability to float and propel through the liquid medium will also make such a craft useful for exploring Titan's methane lakes to perform depth measurements, studying currents and properties of liquid methane.

In [19], it is noted that mini-drones will face challenges in terms of thermal management, which will limit their range and the amount of collected data. Our craft could be used as a mobile mother-ship, which provides a landing zone, long-distance transportation, communication relay, heating, and charging capabilities [20].

4 Methodology

This study investigates the feasibility of screw propulsion for a Titan rover by combining a database of existing vehicles and concepts with a preliminary sizing model. The analysis begins by defining the environmental conditions and mission requirements relevant to the design. The database is then used to identify comparable vehicle configurations, establish reference mass budgets, and collect available performance data. These data support the development of a reduced-order model linking screw geometry to terrain interaction, drive-power demand, and subsystem mass. The resulting estimates are incorporated into an iterative mass budget to obtain a self-consistent vehicle design and assess its compliance with the selected mission requirements.

The database and the sizing model play complementary roles. The database provides reference values and, where sufficiently complete and compatible data are available, potential validation cases. The physical model is used instead to predict the performance of a proposed screw-propelled configuration. However, at present, the database does not contain enough consistent information to establish a calibrated relationship between screw geometry and the mass of the mobility-system. The sizing process therefore combines relationships available in the literature with explicitly stated engineering assumptions. The resulting design estimates should consequently be regarded as preliminary and interpreted in light of both the available data and the simplifying assumptions introduced in the model.

4.1 Reference environment and mission

Titan is adopted as the reference environment for the analysis. Table 1 summarizes the environmental parameters used in the calculations and distinguishes the nom-

inal input of the study from quantities derived from or supported by the literature. The adopted liquid density represents a simplified methane-flotation case and is not intended to reproduce the full compositional and density variability expected in Titan's lakes.

Huygens measurements provide information on the resistance to penetration of the surface at the landing site of the probe [17]. These observations offer useful evidence of local ground conditions, but the terrain model developed here has not been calibrated against them. Instead, the analysis adopts the parameters listed in Table 2 as a representative terrain model under Titan environmental conditions.

The pressure-sinkage coefficients k_c , k_ϕ , and n , together with the shear deformation length K , are taken from the lunar-regolith values reported in Table 1 of the NASA terramechanics white paper [21]. The cohesion, $c = 520$ Pa, and the internal friction angle, $\varphi = 42^\circ$, are retained from the existing model as provisional inputs. The source and compatibility with the remaining terrain parameters still require verification. Both values correspond to the average lunar-soil parameters tabulated by Patel [22] after Carrier et al., which is the source recorded in the sizing code. Therefore, the set of resulting parameters should be understood as an analogue-based terrain representation rather than a calibrated terramechanical description of the surface of Titan.

The rover is assumed to carry a 12 kg science payload and is assessed against a preliminary lander-mass reference of 213 kg. Both values are taken from the growth-predicted column of the TANDEM equipment list [16]. The same source reports a total mass of a system of 344 kg, but this value includes an aeroshell of 131 kg and therefore does not correspond to the lander alone. For this reason, the 213 kg value is used only as a comparative mass benchmark and not as evidence that the proposed screw-propelled rover can necessarily be delivered within this limit.

The preliminary mission requirements further specify a target speed of 0.028 m/s, operation on slopes of up to 20° , and the ability to traverse obstacles approximately 0.10 m high. These values are treated as design requirements for the present analysis rather than as demonstrated capabilities of existing screw-propelled vehicles.

4.2 Database construction and use

The working database contains 24 vehicle and concept records, excluding one synthetic case introduced solely to exercise the software implementation. The dataset includes terrestrial screw-propelled vehicles, laboratory and planetary analogue systems, and comparison concepts based on wheels, tracks, or tensegrity locomotion. Table 3 summarises the current inventory by architecture.

Each database record identifies the vehicle, its operating environment, propulsion architecture, and intended role in the analysis. Where available, numerical fields include total mass, mobility-system mass, payload

mass, power, speed, screw dimensions, and drawbar performance. The database also retains the corresponding source, the definition associated with each reported mass or power value, and an indication of whether a value is reported directly, calculated, or inferred. Missing information is left explicitly blank and is displayed as a dash instead of being assigned a numerical value.

The mobility and payload mass fractions are defined by Eq. (1), where m_{ref} denotes the vehicle mass explicitly selected as a reference for comparison. Meaningful comparison requires the numerator and denominator to refer to the same configuration and to compatible mass definitions. Component masses therefore cannot be used interchangeably with subsystem masses, and values reported as dry, operating, or growth-predicted mass must be distinguished rather than pooled directly.

$$f_{mob} = \frac{m_{mob}}{m_{ref}}, \quad f_{pl} = \frac{m_{pl}}{m_{ref}}, \quad (1)$$

The payload fraction is also sensitive to the selected system boundary. Using the TANDEM growth-predicted values, the 12 kg science payload corresponds to approximately 5.6% of the 213 kg lander mass, but only about 3.5% of the combined 344 kg lander-aeroshell system [16]. The present analysis adopts the lander-only basis. More importantly, the payload mass is specified directly from the mission requirements and is not inferred from the payload fractions observed in terrestrial vehicles.

Table 4 summarises the availability of the main parameters in the working database. These figures indicate only whether a field is populated; they do not imply that the underlying values have been independently verified or are directly comparable. For example, seven records contain a mobility-mass value, but this group includes the incompatible ARCSnake entry. Similarly, only three records provide the complete set of screw dimensions required by the present model, one of which is the full-scale MSA, for which the geometry, test weights, and measured performance were added from the WES report [3]. Database size and field completeness should therefore not be interpreted as the effective sample size of a validated empirical relationship.

At its current level of maturity, the database is primarily used to organise available evidence, identify relevant comparison cases, assess the plausibility of model output, and highlight areas in which additional data are required. As more compatible records become available, their mass fractions and performance data can provide increasingly useful checks on the predicted configurations. However, the final mass of the mobility-system used in the sizing analysis remains based on the sum of the individually modeled components rather than on an empirical regression derived from the database.

4.3 Conceptual design and sizing formulation

4.3.1 Geometry and modeling assumptions

Each screw is defined by five principal geometric parameters: drum diameter D_d , blade height BH , screw

Parameter	Unit	Value	Basis
Atmospheric pressure	Pa	1.5×10^5	Nominal study input
Temperature	K	90	Nominal study input
Atmospheric density	kg/m ³	5.6	Ideal-gas calculation
Gravity	m/s ²	1.352	[14]
Liquid density	kg/m ³	423	Adopted methane case
Near-surface wind speed	m/s	1	Representative value, [15]

Table 1: Nominal environmental inputs. Atmospheric density is calculated from the stated pressure and temperature using a pure-nitrogen ideal-gas approximation. These inputs do not constitute a site-specific environmental specification.

Parameter	Unit	Value	Basis
Cohesive modulus, k_c	N/m ⁿ⁺¹	1400	[21], Table 1
Frictional modulus, k_ϕ	N/m ⁿ⁺²	8.2×10^5	[21], Table 1
Sinkage exponent, n	–	1	[21], Table 1
Cohesion, c	Pa	520	Provisional model input
Internal friction angle, φ	deg	42	Provisional model input
Shear deformation length, K	m	0.018	[21], Table 1
Empirical resistance coefficient, c_{rr}	–	0.10	Provisional model input, as in [14]

Table 2: Adopted terrain inputs. These are lunar analogue and provisional values, not measured Titan terramechanical parameters. The complete set has not been calibrated for screw propulsion.

Architecture	Records
Screw propulsion	14
Screw-wheel hybrid	1
Wheeled	6
Tracked	2
Tensegrity	1
Total, excluding the synthetic example	24

Table 3: Inventory of the current vehicle database. Source eligibility, completeness, and compatibility are assessed separately for each intended use.

length L , number of starts N , and axial lead p . The vehicle is equipped with n_s screws. Here, the lead p represents the axial distance traveled by a single helix over a complete revolution. For a multistart screw, it should therefore be distinguished from the axial spacing between adjacent starts, which is equal to p/N .

$$D_o = D_d + 2BH, \quad D_m = \frac{D_d + D_o}{2}, \quad (2)$$

$$\lambda = \arctan\left(\frac{p}{\pi D_m}\right).$$

The outer diameter D_o , the mean diameter D_m , and the lead angle λ are calculated from the selected geometry according to the relations given above.

The drive torque is estimated using a power-screw analogy through Eq. (3), where μ_{sm} represents the assumed friction coefficient between the screw material and the soil. This formulation provides a convenient reduced-order representation of the relationship between screw geometry and torque, but does not capture the complete mechanics of the screw-soil interaction. As noted by Villacrés et al. [2], the currently

available dynamic models of screw propulsion remain subject to substantial limitations. Therefore, the present analysis uses Eq. (3) as a modeling approximation rather than assuming a universal optimum lead angle or efficiency range.

$$\eta_s = \frac{\tan \lambda}{\tan(\lambda + \rho)}, \quad \rho = \arctan \mu_{sm}, \quad (3)$$

4.3.2 Sinkage and motion resistance

For the terramechanics calculation, each screw is approximated by an equivalent cylindrical contact envelope.

$$b(z) = 2\sqrt{z(D_o - z)}, \quad 0 < z < D_o/2. \quad (4)$$

When the sinkage z remains below the centerline of the cylinder, the corresponding contact width is given by the expression above.

The weight of the vehicle $W = mg$ is initially assumed to be evenly distributed among the n_s screws, giving a reference load per screw of $W_s = W/n_s$. The equilibrium sinkage is then obtained numerically from Eq. (5).

$$\frac{W_s}{b(z)L} = \left(\frac{k_c}{b(z)} + k_\phi \right) z^n. \quad (5)$$

The pressure-sinkage relationship follows the terramechanics formulation described by Li and Bingham [21], while its application to the equivalent cylindrical footprint of a screw is introduced here as a modeling assumption. The equal-load formulation also represents a level-ground approximation. On a uniform slope, the overall normal load would instead scale with $mg \cos \theta$

Field or combination	Populated records, out of 24
At least one total-mass field	21
Mobility mass	7
Payload mass	9
Power	13
Speed	9
Drum diameter, blade height, length and lead together	3
Drawbar coefficient	1

Table 4: Completeness of the working inventory before source and comparability screening. The three records with complete screw geometry are ARCSnake, the screw-wheel hybrid, and the MSA. Only the MSA, a full-scale vehicle with documented test weight, soil, and drawbar performance, provides a validation case for the screw-terrain block (Section 4.5); none of them provides one for the mobility-mass model.

before accounting for the transfer of load between individual screws. Slope calculations based on $W_s = W/n_s$ therefore retain this simplification.

The model requires the calculated sinkage to remain below the height of the blade, $z \leq BH$. Cases exceeding this value are considered outside the adopted operating range. This criterion is intended to limit the extrapolation of the model rather than to establish the accuracy of solutions that remain within the range.

The compaction resistance for each screw is estimated from the pressure-sinkage relationship.

$$R_c = b \left(\frac{k_c}{b} + k_\phi \right) \frac{z^{n+1}}{n+1}. \quad (6)$$

A second empirical estimate is obtained from $R_r = c_{rr}W$, where c_{rr} is treated as a lumped terrain-resistance coefficient. The terrain resistance used in the calculation is then taken as

$$R_{terrain} = \max(n_s R_c, R_r).$$

This convention is introduced to avoid counting the same resistance contribution twice. It should not be interpreted as a general formulation for combining all possible resistance mechanisms. In particular, bulldozing resistance is not explicitly modeled. The implementation also flags cases for which $z/D_o > 0.06$ as a diagnostic indication of relatively high sinkage; this threshold is retained as a provisional screening value rather than as a validated screw-specific limit.

$$F_a = \frac{1}{2} \rho_{atm} C_d A_f v_{rel}^2, \quad (7)$$

Aerodynamic resistance is calculated using the conventional drag expression given above. The relative air speed is approximated by the specified wind speed, while the frontal area is estimated as $A_f = n_s D_o h_f$ using an effective height $h_f = 1.2$ m. Combining aerodynamic drag with terrain resistance, acceleration, and the gravitational component along the slope gives the required propulsive force in Eq. (8).

$$F_{req} = ma + W \sin \theta + R_{terrain} + F_a. \quad (8)$$

The aerodynamic force is assumed to act directly opposite the direction of travel. Crosswind effects are not

included and would require a separate directional stability and loading analysis.

4.3.3 Available thrust

The available axial thrust is estimated by combining a Mohr-Coulomb shear limit with a slip-dependent mobilisation term, as expressed in Eq. (9). The contact area is approximated as $A_c = b(z)L$, while the shear displacement is represented by $j = sL$ for the selected slip ratio s .

$$F_{av} = n_s [A_c c + W_s \tan \varphi] \times \left[1 - \frac{K}{j} \left(1 - e^{-j/K} \right) \right]. \quad (9)$$

Li and Bingham [21] describe an analogous contact-patch formulation for smooth wheels. In the present model, the resulting shear force is applied directly as axial screw thrust. This extension simplifies the actual mechanics of screw locomotion because it does not resolve blade engagement, lateral forces, or interactions between adjacent screws.

No Titan-specific drawbar coefficient is assumed. Terrestrial drawbar measurements may eventually provide useful comparison or calibration cases, provided that associated terrain properties, screw geometry, slip ratio, and force definitions are sufficiently documented. Particular care is required when distinguishing net drawbar pull from gross thrust, since subtracting motion resistance from an already net force would count the same effect twice.

To account for the difference between the shear capacity of the soil and the thrust actually developed by a rotating screw, the revised model introduces a screw traction efficiency $\kappa \leq 1$, which scales the Mohr-Coulomb limit. Where a terrain case includes a measured drawbar coefficient μ_{db} , the available thrust is additionally limited to $\mu_{db}W_s$:

$$F_{av} = n_s \min\{\kappa [A_c c + W_s \tan \varphi], \mu_{db}W_s\} \times \left[1 - \frac{K}{j} \left(1 - e^{-j/K} \right) \right]. \quad (10)$$

For $\kappa = 1$ and in the absence of a measured limit, Eq. (10) reduces to Eq. (9). The calibration of κ against the MSA test data is discussed in Section 4.5.

Within these assumptions, feasibility of the traction requires the predicted available thrust to satisfy

$$F_{av} \geq F_{req}.$$

4.3.4 Kinematics, torque and power

$$v_i = \frac{v}{1-s}, \quad \omega = \frac{2\pi v_i}{p}. \quad (11)$$

For a prescribed ground speed v and a slip ratio $0 \leq s < 1$, the ideal advance speed and the angular velocity of the screw are obtained from the screw lead by the relations given above.

The corresponding torque per screw and the total electrical drive power are then calculated using Eq. (12). The efficiency of the drivetrain η_{dt} accounts for the assumed losses between the electrical input and the screw output. Slip is treated as an imposed operating parameter rather than being solved as an equilibrium outcome of the terrain-interaction model. In addition, the model reports the equilibrium slip s_{eq} , defined as the slip at which $F_{av}(s) = F_{req}$ and obtained by bisection on the monotonic mobilisation term of Eq. (10). This value represents the operating point expected for the vehicle, whereas the drive power continues to be sized at the imposed design slip, which is a conservative choice. If no slip satisfies the equality, the configuration is classified as traction limited.

$$T = \frac{F_{req}}{n_s} \frac{p}{2\pi\eta_s}, \quad P_{el} = \frac{n_s T \omega}{\eta_{dt}}, \quad (12)$$

For a selected nominal motor speed ω_m , the required reduction ratio G and motor torque τ_m are subsequently calculated from the screw rotational speed. The motor-torque estimate includes the assumed gearbox efficiency η_g and a torque margin $\gamma_T = 2$.

$$G = \frac{\omega_m}{\omega}, \quad \tau_m = \frac{\gamma_T T}{G\eta_g}, \quad (13)$$

4.3.5 Mass estimation and closure

The mobility-system mass is constructed from the running gear, drive modules, and drive electronics. Running-gear mass is estimated using Eq. (14), with effective mass-per-area coefficients σ_d and σ_b for the drum and blades, respectively, and a support allowance f_b .

$$m_{rg} = n_s \left[\pi D_d L \sigma_d + \frac{L}{p} N \frac{\pi(D_o^2 - D_d^2)}{4} \sigma_b \right] (1 + f_b). \quad (14)$$

The drum contribution represents the cylindrical lateral surface, while the blade contribution is based on the projected annular area per turn. The calculation therefore does not explicitly size features such as drum end closures, seals, internal reinforcement, or the developed

surface of the helicoid. The resulting value is intended as a preliminary component-level mass estimate rather than a structural design.

The baseline sizing model assigns a fixed drive-module mass of 1.5 kg per screw and an additional 0.8 kg per screw for drive electronics.

Power-system mass is estimated from the combination of the housekeeping load and the duty-cycled mobility load. For a drive duty cycle δ , the nominal continuous source power is calculated as

$$P_{src} = P_{house} + \delta P_{el}.$$

The corresponding source and battery masses are then obtained from the assumed source-specific power and battery-specific energy. The adopted battery value of 100 Wh/kg follows the warm-battery assumption reported in the Titan Aerial Daughtercraft study [15]. A separate allowance is included for power-management hardware.

$$m_{src} = \frac{P_{src}}{q_{src}}, \quad m_{bat} = \frac{\max(0, P_{house} + P_{el} - P_{src}) t_d}{e_{bat}}. \quad (15)$$

This simplified model does not explicitly account for battery aging, depth-of-discharge limits, or charge and discharge losses. Similarly, linear scaling of source mass with required power is used as a conceptual sizing relationship and does not replace the eventual selection and sizing of an actual power unit.

The remaining non-mobility subsystems are represented as fractions f_i of the un-margined vehicle mass. These fractions are provisional design allocations rather than established Titan-system mass fractions. The vehicle dry and total masses are calculated using Eq. (16). In this formulation, m_{dry} denotes the un-margined assembled mass, including the payload and power-system hardware, while f_m represents the system-level mass margin. The subsystem fractions are assumed to describe mutually exclusive allocations and must satisfy $\sum_i f_i < 1$.

$$m_{dry} = \frac{m_{pl} + m_{mob} + m_{eps}}{1 - \sum_i f_i}, \quad m_{tot} = m_{dry} (1 + f_m). \quad (16)$$

Mass closure is solved iteratively. Starting from an initial estimate of total vehicle mass, the model recalculates sinkage, terrain resistance, drive power, and power-system mass before updating the overall vehicle mass. A relaxation factor of 0.5 is applied between iterations, and convergence is reached when the relative change in mass falls below 0.1% or when the numerical stopping limit is reached.

Screw geometry remains fixed during an individual closure calculation and is varied only between candidate designs. Under the current mass model, the running-gear and drive-module masses therefore remain constant

within a given run, while the main load-dependent feedback occurs through the power subsystem. Failure of the numerical iteration to converge is reported as an unresolved sizing case rather than being interpreted as evidence that the underlying physical configuration is necessarily infeasible.

4.3.6 Flotation

Flotation is assessed by comparing the maximum volume displaced by the screw drums with the liquid volume required to support the estimated vehicle mass. The two quantities are defined in Eq. (17).

$$V_{drums} = n_s \frac{\pi D_d^2}{4} L, \quad V_{req} = \frac{m_{tot}}{\rho_{liq}}. \quad (17)$$

This calculation assumes that the drums are sealed and preserve their external volume. The contribution of the screw blades and any additional buoyant hull volume is neglected. As expected from the static buoyancy balance, gravity cancels from the comparison, while the assumed liquid density remains an explicit input.

A configuration satisfying

$$V_{drums} \geq V_{req}$$

has sufficient maximum drum displacement according to this simplified static calculation. Passing this check does not, however, establish adequate freeboard, trim, or stability, and equality would provide no reserve displacement. Conversely, a design that fails the drum-only condition would require additional buoyant volume elsewhere in the vehicle. A complete assessment of amphibious operation would also need to consider hydrodynamic resistance, propulsion in liquid, sealing, trim, and stability.

4.3.7 Feasibility and model assessment

The assessment distinguishes three different aspects of the design: numerical mass convergence, compliance with the selected design requirements, and validity of the underlying physical model. Table 6 summarises the preliminary screening criteria applied to candidate configurations.

These criteria do not all have the same evidential status. The mass limit is an adopted mission benchmark, while traction is a conditional prediction of the current terrain-interaction model. Other criteria, including the relative-sinkage threshold, mobility-fraction range, and obstacle indicator, are engineering screens introduced to identify potentially problematic configurations rather than experimentally established performance boundaries.

The wind-overturning calculation compares the restoring moment $Wb_t/2$ with the overturning moment $F_a h_{cg}$. The assumed support width is $b_t = n_s D_o$, and h_{cg} denotes the estimated centre-of-gravity height. This provides a first-order stability indicator but does not resolve the actual vehicle support polygon, slope-induced

tipping, or dynamic load transfer. It should therefore be interpreted as a screening calculation rather than a complete rollover analysis.

Physical validation of the mobility model requires an independent and internally consistent test case for which vehicle geometry, total mass, terrain properties, slip, force, and power are all known under the same operating conditions. The measured and predicted quantities must also share consistent definitions, particularly for force and power.

The current database does not yet contain a complete real-vehicle dataset that satisfies these requirements for the implemented reconstruction procedure. The synthetic case can be used to verify that the software and numerical workflow execute correctly, but it cannot establish the physical validity of the underlying model. The trafficability report of the MSA [3] nevertheless documents the rotor geometry, test weights, soil conditions, and measured performance of a full-scale screw-propelled vehicle. These data are sufficient to assess the screw-terrain block and the behavior of the closure loop, although they do not allow the mobility-mass model to be verified. This assessment is presented in Section 4.5.

The present framework is therefore intended primarily for preliminary design exploration. It allows candidate screw geometries to be sized consistently, enables their implications for mobility, power, and vehicle mass to be examined, and helps identify the assumptions that have the greatest influence on the predicted design. At this stage, it does not constitute a validated demonstration of screw-propelled mobility on Titan.

Further development should focus on improving the terrain analogue, calibrating screw-specific force and power relationships, refining component-level and structural mass models, and incorporating physical effects that are currently neglected, including lateral motion, dynamic sinkage, repeated terrain passes, screw-screw interaction, and operation in liquid. Sensitivity analysis should then be used to determine how strongly these uncertainties affect the resulting design conclusions, while maintaining a clear distinction between assumed parameter ranges and statistically derived confidence intervals.

4.4 Case Study

The baseline configuration is a two-screw vehicle with the geometry and operating assumptions listed in Table 7, the mission requirements of Section 4.1, and the coefficients of Table 5. The geometry corresponds to the smallest configuration of the trade space discussed in Section 5.5 that satisfies all the screens of Table 6, including the drum-only flotation screen, on the lunar-proxy soil. It should therefore be regarded as an outcome of the trade-off rather than as an independent design choice. The resulting lead angle at the mean diameter is 12.8° , which corresponds to a power-screw efficiency of 0.24 for the assumed soil-metal friction coefficient of 0.6.

Coefficient or allocation	Value	Status
Drum coefficient, σ_d	12.0 kg/m ²	Provisional
Blade coefficient, σ_b	29.1 kg/m ²	Provisional
Support allowance, f_b	0.10	Assumed
Drive module, per screw	1.5 kg	Provisional
Drive electronics, per screw	0.8 kg	Assumed
Structure fraction	0.097	Provisional
Thermal fraction	0.075	Provisional
Data-handling fraction	0.030	Provisional
Navigation fraction	0.014	Provisional
Communications fraction	0.045	Provisional
Harness fraction	0.021	Provisional
System mass margin, f_m	0.156	Assumed
Source-specific power, q_{src}	2.44 W/kg	Provisional scaling input
Battery-specific energy, e_{bat}	100 Wh/kg	Warm-battery assumption, [15]
Power-management allowance	0.15	Assumed
Drivetrain efficiency, η_{dt}	0.472	Provisional
Gearbox efficiency, η_g	0.59	Provisional
Soil-metal friction, μ_{sm}	0.60	Assumed
Screw traction efficiency, κ	1.0; 0.35 to 0.65	Uncorrected; calibrated on the MSA

Table 5: Coefficients retained from the current sizing model. Provisional values require verification or calibration before they can support a final design; they are not presented as measured properties of Titan hardware.

Screen	Criterion	Interpretation
Mass	$m_{tot} \leq 213$ kg	Adopted lander benchmark
Relative sinkage	$(D_o/2)/z \geq 6$	Assumed screening threshold
Traction	$F_{av} \geq F_{req}$	Conditional model prediction
Wind overturning	Moment ratio ≥ 2	Simplified wind-load screen
Mobility fraction	$0.15 \leq m_{mob}/m_{tot} \leq 0.60$	Assumed plausibility interval
Obstacle indicator	$0.5D_o \geq 0.10$ m	Unvalidated wheel analogy
Drum displacement	$V_{drums} \geq V_{req}$	Static buoyancy screen

Table 6: Preliminary design screens. The mobility-fraction interval is not a statistically established database bound, and the obstacle indicator does not demonstrate screw-rover climbing capability.

Parameter	Symbol	Value
Number of screws	n_s	2
Drum diameter	D_d	0.55 m
Blade height	BH	0.08 m
Outer diameter	D_o	0.71 m
Screw length	L	1.10 m
Axial lead	p	0.45 m
Number of starts	N	1
Lead angle at D_m	λ	12.8°
Design slip	s	0.30
Motor nominal speed	ω_m	5000 rpm
Drive duty cycle	δ	0.15
Drive session	t_d	4 h

Table 7: Baseline geometry and operating assumptions. The geometry results from the trade space of Section 5.5.

The closure loop converges in six iterations from an initial estimate of 200 kg. Table 8 reports the converged mass and power budget. The running gear accounts for 38% of the total mass and the drive modules for 2.3%. The resulting mobility mass fraction of 0.40 falls within the 0.16 to 0.49 range of the screw-propelled vehicles in the database, but also within the 0.15 to 0.47 range of the

wheeled rovers; for this reason, the database alone does not support conclusions for or against screw propulsion on the basis of mass. The power source is sized at 62 W, of which 55 W correspond to housekeeping loads and 7 W to the duty-cycled drive, while a 4 h drive session at full drive power determines the 1.5 kg battery. At 0.028 m/s, the drive requires 445 Wh per kilometre, and the assumed 15% duty cycle corresponds to approximately 360 m of traverse per Earth day.

Table 9 summarises the performance indicators and the outcome of each screen. The static sinkage is 1.9 mm, the contact pressure is 1.6 kPa, and the ratio between outer radius and sinkage is 182, well above the threshold of 6. The terrain resistance is governed by the empirical coefficient, which gives 26.7 N, rather than by Bekker compaction, which gives 0.2 N. The near-surface wind on Titan is weak, approximately 1 m/s [15]; despite the dense atmosphere, it produces an aerodynamic drag of only 5.7 N and a wind-overturning ratio of 55. Wind is therefore not a design driver for a surface vehicle of this size. The required thrust of 126 N on the design slope is met with an uncorrected margin of 2.45, and the vehicle is predicted to operate at an equilibrium slip of 1.8%, well below the 30% design slip used for

Item	Mass, kg	Power, W
Science payload	12.0	15.0
Running gear	75.0	
Drive modules and electronics	4.6	44.8
Power source, MMRTG class	25.3	61.7
Battery	1.5	
Power management	4.0	
Structure	16.5	
Thermal	12.8	20.0
Data handling	5.1	20.0
Navigation	2.4	
Communications	7.7	
Harness	3.6	
Dry mass	170.5	
System margin (15.6%)	26.6	
Total	197.1	

Table 8: Converged baseline budget on the lunar-proxy soil. The drive power is the electrical input at the design point (20° slope, 0.028 m/s, 30% slip). The source of the multi-mission radioisotope thermoelectric generator (MMRTG) class with 2.44 W/kg [14] is sized for the housekeeping loads plus the duty-cycled drive; the 20 W reported for data handling is the avionics allocation.

power sizing. The drums displace 0.52 m³, compared with the 0.47 m³ required to float the converged mass in liquid methane, which corresponds to a margin of 1.12 without any contribution from the blades or from a hull.

Indicator	Value	Screen
Total mass	197 kg	≤ 213 kg, pass
Static sinkage	1.9 mm	$D_o/2z = 182$, pass
Contact pressure	1.63 kPa	
Thrust required, 20°	126 N	
Thrust available, $\kappa = 1$	307 N	margin 2.45, pass
Equilibrium slip, s_{eq}	0.018	
Screw torque, per screw	18.9 N m	
Screw speed	5.3 rpm	
Motor torque with margin	0.07 N m	at $G = 938$
Mobility mass fraction	0.40	0.15 to 0.60, pass
Obstacle indicator, $0.5D_o$	0.36 m	≥ 0.10 m, pass
Wind overturning ratio	55	≥ 2, pass
Drum displacement	0.52 m ³	≥ 0.47 m ³ , pass

Table 9: Baseline performance and screening outcome on the lunar-proxy soil.

4.5 Validation

The validation is carried out at two levels. First, the screw-terrain block is compared directly with the measured data; second, the closure loop is used to design a new vehicle to the requirements of an existing one. Both comparisons are based on the full-scale MSA, as documented by the WES [3]. The report provides the rotor geometry (drum diameter of 26 in, 31 in over the helix, 129.5 in in contact with the ground, and 48 in lead), the empty and loaded test weights (2860 and 3954 lb), the ground pressure at 3 in of penetration, the rut depth on sand, the maximum towing force on sand and on

wet clay, and the slopes climbed and failed. To the authors' knowledge, these are the only published data for a full-scale screw-propelled vehicle in which geometry, weight, soil conditions, and performance are reported together. The helix is assumed to be single-start, and the soil is represented by the dry-sand parameter set tabulated by Patel [22], since the report provides cone index values rather than Bekker parameters.

Screw-terrain block. Table 10 compares the predictions of the model with the reported data. The equivalent-cylinder contact model reproduces the ground pressure at 3 in of penetration to within 16% for both test weights; since the reported value is itself a computed ground pressure, this comparison verifies the contact-area convention rather than the soil model. The static Bekker sinkage on the dry-sand set is 37 mm, compared to a first-pass rut of 142 mm, and decreases to 11 mm with the stiffer parameter set usually attributed to Wong. The report describes digging and bulldozing on sand, neither of which is included in the static model. On sand, the model therefore underestimates the rut by a factor of 4 to 13, and the sinkage screen of Table 6 should be interpreted as a screen on static flotation only.

Quantity	Model	Report	Note
Ground pressure, empty	4.15 kPa	3.59 kPa	+16%
Ground pressure, loaded	5.74 kPa	4.96 kPa	+16%
Sinkage, sand, empty	37 mm	142 mm	rut
Traction, sand	0.68	0.24	towing
		0.40 to 0.46	slopes
Traction, wet clay	–	0.107	$s = 0.28$

Table 10: Screw-terrain block compared with the MSA tests [3]. Ground pressure at 3 in of penetration; the rut is measured after one pass; traction is expressed as a coefficient of the test weight. The model traction is the uncorrected Mohr-Coulomb limit of Eq. (9) on the dry-sand set of [22]; the slope-derived range adds the rolling resistance coefficient to the tangents of the 18° slope climbed and of the 21° slope failed.

The comparison of traction provides the most relevant result of the validation. The Mohr-Coulomb limit gives a traction coefficient of 0.68 for the loaded vehicle. The report instead measured a towing force of approximately 24% of the test weight at full throttle and 0.5 mph, and the loaded vehicle climbed a 18° sand slope but not a 21° one, which, including the rolling resistance coefficient, bounds the usable coefficient between 0.40 and 0.46. The ratio between measured and modeled traction, which corresponds to the screw traction efficiency κ of Eq. (10), is therefore 0.36 from the towing test and 0.61 to 0.70 from the slope tests. On wet clay, the maximum towing force was 425 lb, or 10.7% of the test weight, at a slip of about 28%, and decreased at higher slip. This behavior is qualitatively consistent with the slip-mobilisation term, but a quantitative comparison is not possible because the report provides rating

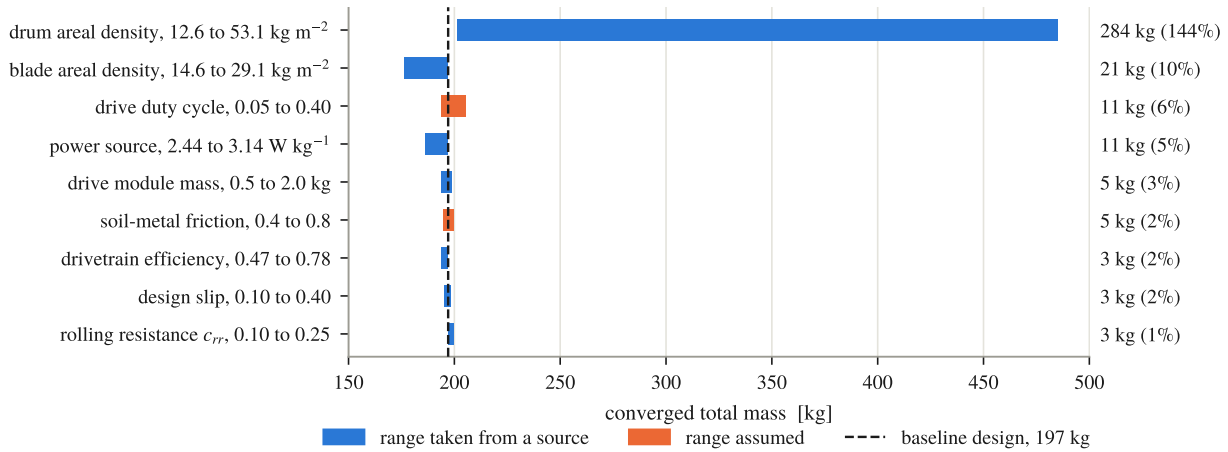


Fig. 2: One-at-a-time sensitivity of the converged total mass. Each bar shows the interval of converged total mass obtained when the input on the left is varied across the indicated range, with all other inputs at their baseline values. Blue bars denote ranges taken from the literature and orange bars ranges assumed in the present study; the dashed line marks the 197 kg baseline, and the span of each interval is reported on the right in kilograms and as a percentage of the baseline.

cone index values rather than shear parameters. The values $\kappa = 0.65$ and $\kappa = 0.35$ are carried forward into the Titan analysis.

Closure loop. The complete loop was then applied as if a new vehicle had to be designed to the requirements of the MSA: a payload of 496 kg (the difference between loaded and empty test weights), a speed of 0.72 m/s (the 1.6 mph average on sand), an 18° sand slope, and the MSA rotor geometry. An engine-driven mass model was used, whose coefficients are all assumed (0.4 kW/kg engine, 25 kg drive module per rotor, hull and frame equal to 30% of the dry mass, and no thermal or avionics allocation). Table 11 compares the converged vehicle with the vehicle that was built. The loop converges in ten iterations at 1170 kg, approximately one third lighter than the loaded MSA and about half of its empty mass, since a rover structural fraction does not represent a welded aluminium hull with a cab; the converged mass should therefore not be interpreted as a prediction. The two screens are more informative. The drum-only flotation screen is satisfied with a margin of 1.9, which is consistent with a vehicle that operated at 8 mph on water. The traction screen indicates that the vehicle can climb the 18° slope with $\kappa = 0.65$ (margin 1.19) but not with $\kappa = 0.35$ (margin 0.64). Since the real vehicle did climb this slope, the towing figure was most likely limited by both power and traction, and the slope-derived value is considered the more representative calibration.

A validation of this type verifies the structure of the model and identifies where the model is optimistic on a granular ground. It does not validate the performance on Titan, and therefore the Titan results are presented for the calibrated range of κ as well as for the uncorrected value.

Quantity	Loop	MSA
Total (loaded) mass, kg	1170	1794
Empty mass, kg	674	1297
Running gear, kg	205	–
Hull and frame, kg	351	–
Drum displacement / volume to float	1.93	float
Drawbar margin, 18°, $\kappa = 1$	1.83	
Drawbar margin, 18°, $\kappa = 0.65$	1.19	climbed
Drawbar margin, 18°, $\kappa = 0.35$	0.64	climbed

Table 11: Closure loop applied to the design of a new vehicle to the MSA requirements, compared with the vehicle that was built [3]. All mass coefficients of the loop are assumed for an engine-driven terrestrial vehicle.

5 Results and Discussion

5.1 Drivers of the vehicle mass

Fig. 2 reports a one-at-a-time sensitivity analysis of the converged total mass. Each bar corresponds to one uncertain input, which is varied throughout the range indicated in its label, while all other inputs remain at their baseline values; the horizontal extent of the bar is the resulting interval of converged mass, and the dashed line marks the 197 kg baseline. Blue bars denote ranges taken from the literature, whereas orange bars denote ranges assumed in the present study, and the span of each interval is reported on the right-hand side. The inputs are ordered by decreasing influence.

The drum shell areal density is by far the most influential input. Across the range of 12.6 to 53.1 kg m⁻² for the micro-rover wheels reported by Patel [22], the converged mass varies from 201 to 485 kg, a span of 144% of the baseline. The blade areal density follows with 10%, the drive duty cycle with 6%, and the specific power of the source with 5%. All the remaining inputs, including the drive-module mass, the soil-metal friction,

the drivetrain efficiency, the design slip, and the rolling resistance coefficient, affect the result by 3% or less. In a screw-propelled vehicle, the mobility mass is therefore carried by the running gear rather than by the actuators, and the quantity that most requires measurement is the mass per unit area of the drum and of the helix, rather than a regression of motor mass.

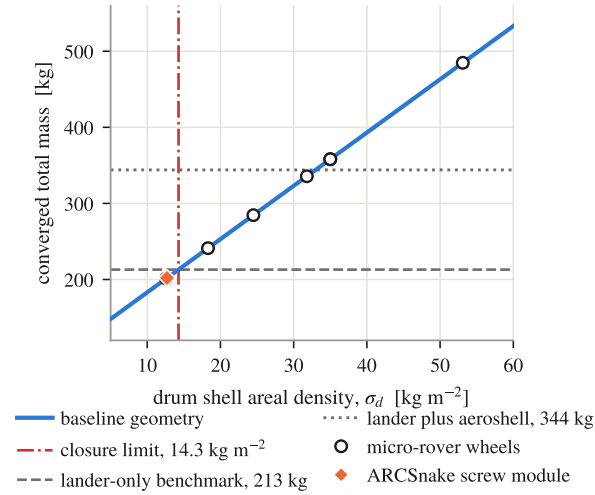


Fig. 3: Converged total mass of the baseline design as a function of the drum shell areal density σ_d . The horizontal lines are the 213 kg lander-only and 344 kg lander-plus-aeroshell references [16]; the vertical line is the closure limit, at which the converged mass equals the lander-only reference. Open circles are micro-rover wheels [22], from left to right ELMS (body text), Marsochod, ELMS (Table 68), Sojourner and Shrimp, and Nanokhod; the Crab wheel (12.6 kg m^{-2}) is hidden by the diamond, which is the ARCSnake screw module including its drive (12.7 kg m^{-2}) [13].

Given this dominance, the drum areal density is treated more usefully as a requirement than as an input. Fig. 3 reports the converged total mass as a function of σ_d , obtained by repeating the closure calculation for each value with all other inputs at the baseline. The horizontal lines indicate the two mass references and the vertical line marks the closure limit, defined as the areal density at which the converged mass reaches the 213 kg lander-only benchmark; configurations to its left satisfy the mass screen, whereas configurations to its right do not. The markers position the available evidence on the curve. The closure limit is 14.3 kg m^{-2} . Only the ARCSnake screw module, at 12.7 kg m^{-2} with its drive and electronics included [13], and the lightest micro-rover wheel, at 12.6 kg m^{-2} , fall below it. The micro-rover wheels are 10 to 14 cm components machined from billet, whereas a 0.55 m drum would be manufactured as a rolled shell supported by ribs; the comparison should therefore be regarded as an upper bound rather than as a contradiction. Nevertheless, the available margin with respect to the only measured screw module is small, and the drum areal density is stated here explicitly as a re-

quirement on the structural design.

5.2 Flotation and sinkage

Contact pressure and sinkage scale with the vehicle weight. The baseline of 197 kg operates at 1.6 kPa with a sinkage of 2 mm on lunar-proxy soil, compared to 7.0 kPa and 8.3 mm for the 40 kg wheeled Titan rover of Genta and Genta on the same soil [14], and with the recommended and allowable lunar design values of 1.4 and 8 kPa [23]. The same vehicle would sink 7.4, 2.2, and 1.9 mm under Earth, lunar, and Titan gravity, with contact pressures of 6.1, 1.8, and 1.6 kPa, respectively. Flotation on soft ground is therefore not a limiting factor on Titan for a vehicle of this class, and the drum-only displacement screen is satisfied with a margin of 1.12. Neither result, however, demonstrates an amphibious capability: the sinkage model underestimates the rut produced by a rotating screw on sand (Section 4.5), and freeboard, trim, stability, and hydrodynamic drag are not modeled.

5.3 Traction on the design slope

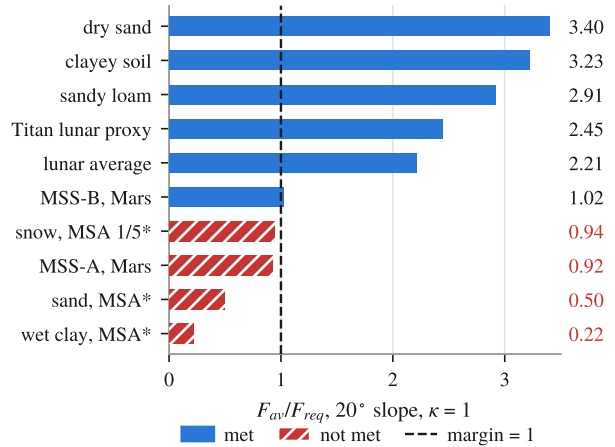


Fig. 4: Ratio between available and required thrust of the baseline design on the 20° design slope, computed with the uncorrected model ($\kappa = 1$) for every terrain case with Bekker parameters. The dashed line marks a ratio of one; hatched red bars do not meet the requirement, and the value of each ratio is reported on the right. Cases marked with an asterisk are the only ones with a measured screw drawbar coefficient, all from the MSA program [2, 3]; their Bekker parameters are assumed.

Fig. 4 compares, for each terrain case with Bekker parameters, the ratio between the available and the required thrust of the baseline design on the 20° design slope, computed with the uncorrected model. Ratios greater than one, to the right of the dashed line, indicate that the traction requirement is met, and hatched red bars indicate that it is not. On the lunar-proxy soil, the ratio is 2.45, and the vehicle is predicted to operate at a slip of 1.8%. On the cohesive-frictional soils, the ratio ranges between 2.2 and 3.4, whereas it is close to one on the Mars simulants and on snow, and falls to 0.50 and 0.22

in the two cases that carry the full-scale MSA drawbar measurements on sand and wet clay. The requirement is therefore not met on any of the soils for which a screw drawbar coefficient has been measured, while the larger ratios are obtained on soils for which no screw test data are available.

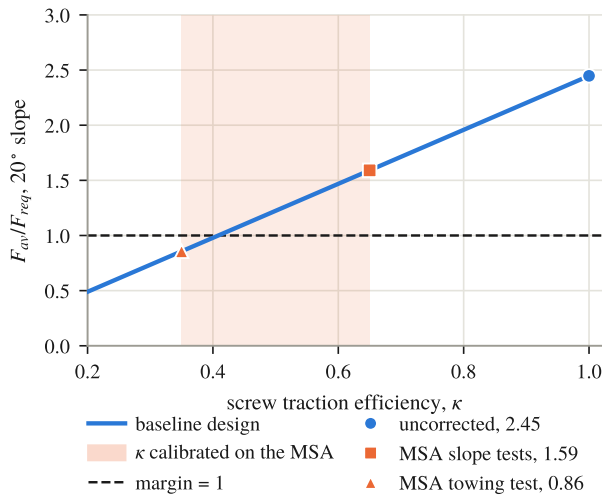


Fig. 5: Ratio between available and required thrust of the baseline design on the 20° slope of the lunar-proxy soil as a function of the screw traction efficiency κ . The dashed line marks a ratio of one, and the shaded band is the range of κ calibrated on the MSA in Section 4.5. The markers identify the uncorrected model and the two calibrated values, with the corresponding ratios reported in the legend.

Fig. 5 shows how this ratio varies with the screw traction efficiency κ on the lunar-proxy soil. The solid line is the ratio between the available and required thrust, the dashed line marks a ratio of one, and the shaded band corresponds to the range of κ calibrated on the MSA. The three markers identify the uncorrected model ($\kappa = 1$) and the two calibrated values derived from the slope tests ($\kappa = 0.65$) and from the towing test ($\kappa = 0.35$). With slope-derived calibration, the ratio decreases to 1.59, and the equilibrium slip increases to 3.6%, while with the towing-derived calibration, the ratio decreases to 0.86 and no slip satisfies the thrust requirement. Expressed as the largest slope that the force balance can sustain, the uncorrected model does not predict a traction limit on this cohesive proxy soil, since the available thrust exceeds the vehicle weight, whereas the calibrated values give 39° and 16°, respectively. The converged mass does not change with κ , because the thrust is sized from the requirement rather than from the available ceiling. Traction on the design slope, rather than flotation or mass, is therefore the aspect of the design most affected by the terrain uncertainty, and the slope requirement should be the first to be revisited when a Titan-relevant soil parameter set becomes available.

5.4 Effect of gravity

For a purely frictional soil, the weight cancels between the required and the available force, and the traction-limited slope is independent of gravity. In the present model, this cancellation is only partial, because the cohesive term $A_c c$ in Eq. (10) does not scale with weight and therefore represents a larger share of the available thrust at low gravity. For the same vehicle on the lunar-proxy soil, the available traction coefficient increases from 0.93 under Earth gravity to 1.12 under lunar gravity and 1.15 under Titan gravity. In the wet-clay case, where the thrust is limited by the measured drawbar coefficient, the coefficient does not change with gravity, and the slope that the vehicle can sustain remains at about 2°. Whether the cohesion effect applies on Titan depends on a surface cohesion that has been measured only at the Huygens landing site [17]. The same method can be applied to lunar and Martian cases by changing the environment and terrain inputs; the results obtained on the Mars simulants in Fig. 4 already indicate that the design slope would be marginal in those conditions.

5.5 Feasibility box

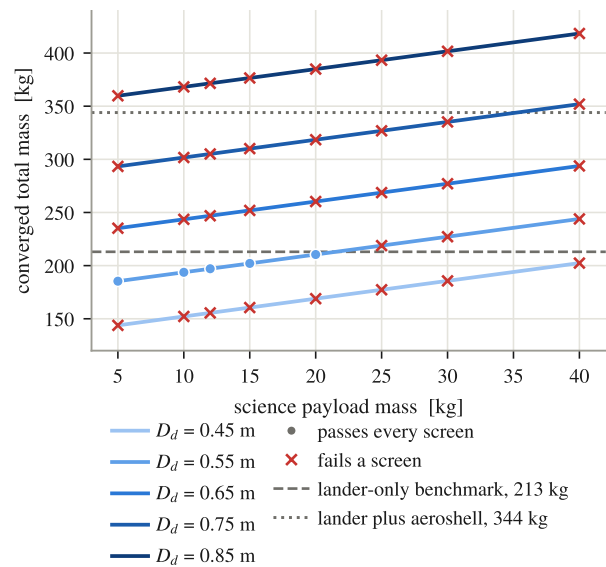


Fig. 6: Feasibility box of the Titan screw-propelled rover. Converged total mass as a function of science payload mass for five drum diameters, with blade height, screw length, and lead scaled in the proportions of the baseline, so that the 0.55 m line passes through the baseline design. Circles satisfy every screen of Table 6 and crosses fail at least one; the dashed and dotted lines are the 213 kg and 344 kg mass references [16].

Fig. 6 presents the feasibility box of the design. Each line corresponds to a drum diameter between 0.45 and 0.85 m, with blade height, screw length, and lead scaled in the same proportions as the baseline, and reports the converged total mass as a function of the science payload mass. Circles denote configurations that satisfy all

the screens in Table 6, and crosses denote configurations that fail at least one of them. Under the lander-only benchmark, the feasible region is narrow: only the 0.55 m drum satisfies every screen, for payloads up to 20 kg. The 0.45 m drum fails the drum-only flotation screen at every payload, and drums of 0.65 m and above exceed the mass benchmark. With respect to the 344 kg lander-plus-aeroshell reference, the 0.65 m drum would remain feasible up to 40 kg of payload and the 0.75 m drum up to 30 kg. Since the benchmark is taken from a comparator study and is not derived from an entry, descent, and landing (EDL) analysis of a screw-propelled rover, the feasibility box should be interpreted as an indication of the shape of the constraint rather than as a conclusive assessment of the concept.

5.6 Limitations

The Titan soil is represented by a lunar proxy, adopted after [14], which constitutes the largest single source of uncertainty in the study. The Bekker parameters of the MSA terrain cases are assumed or borrowed from other soils, since the report provides only cone index values. The model does not include lateral drift and sideslip [10, 11], the transition to wheel-like rolling on hard ground [13], dynamic and slip-induced sinkage, bulldozing, multi-pass effects, interaction between screws, hydrodynamic drag and thrust in liquid, or the two failure modes observed on Earth, namely immobilisation by adhesive mud on ground that is otherwise firm enough to support the vehicle [2, 3] and stall in saturated sand [10]. The drive is sized with a fixed module mass and a 938:1 reduction from a 5000 rpm motor; direct drive at screw speed, as adopted in the Titan rover of [14], represents a valid alternative that is not sized by the present model. The thermal fraction is derived from a Mars rover budget and is the least transferable coefficient in Table 5 for operations at 90 K with a radioisotope source. Finally, the MSA data were extracted from the digitised copy of the WES report and should be verified against the scanned pages before being used further.

6 Conclusion

This paper has presented a Phase 0/A sizing method for screw-propelled planetary rovers, in which a subsystem mass and power budget is coupled with a reduced-order screw-terrain model within an iterative closure loop, supported by a vehicle database in which each value is associated with its definition, evidence level, and source. The comparison with the full-scale MSA trafficability tests shows that the contact model reproduces the reported ground pressure within 16%, that the static sinkage model underestimates the rut produced on sand, and that the Mohr-Coulomb traction limit overestimates the measured drawbar pull by a factor of 1.5 to 3. This difference has been expressed as a screw traction efficiency between 0.35 and 0.65 and carried into the planetary case. When applied to the design of a new vehicle to the MSA requirements, the loop converges in the correct mass class, and its flotation and traction

screens are consistent with the observed behavior of the real vehicle.

For a Titan reference mission with a 12 kg payload on a lunar-analogue soil, the method converges on a two-screw vehicle of 197 kg with 45 W of electrical drive power, within the 213 kg lander benchmark of a recent Titan rover study. The areal density of the running gear dominates the mass uncertainty, and closure within the benchmark requires a drum shell below 14 kg per square metre, a value only just met by the single measured screw module. Flotation is not a limiting factor on Titan, with a contact pressure of 1.6 kPa. Traction on the 20° design slope becomes the governing performance constraint once the terrestrial calibration is applied, with a drawbar margin between 0.9 and 1.6, and low gravity improves the traction-limited slope only through the cohesion term. The method does not demonstrate that screw propulsion is preferable to wheeled or tracked mobility on Titan, since the database ranges of mobility mass fraction overlap and the terrain is represented by a proxy. It does, however, quantify the cost of a screw-propelled option and identify its most sensitive aspects, which is the information required for a Phase 0 trade-off.

The measurements that would most reduce the uncertainty are, in order of priority, the mass per unit area of a flight-representative drum and helix, a screw-specific drawbar and torque test on a Titan-relevant granular simulant at low contact pressure, and the terramechanical parameters of the surface of Titan. Future developments should include lateral and dynamic sinkage effects, an operating-point slip for power as well as for traction, and a model of operation in liquid before any amphibious capability is claimed.

The future work under that topic includes more thorough modeling of the interaction between screws and soil, using the specialized software; study of the hydrodynamical characteristics of the designed mobility system in liquid methane; further optimisation and finalisation of the proposed design.

References

- [1] F. B. Stevens. *The First Steam Screw Propeller Boats to Navigate the Waters of Any Country*. Engineering Press, 1893. URL <https://www.museum.state.il.us/RiverWeb/landings/Ambot/Archives/stevens/index.html>. Reprinted from The Stevens Indicator, volume 10, number 2, April 1893.
- [2] J. Villacrés, M. Barczyk, and M. Lipsett. Literature review on Archimedean screw propulsion for off-road vehicles. *Journal of Terramechanics*, 108:47–57, 2023. doi: 10.1016/j.jterra.2023.05.001.
- [3] U.S. Army Engineer Waterways Experiment Station. Trafficability tests with the Marsh Screw Amphibian on coarse-grained and fine-grained soils. Technical Report 3-641, U.S. Army Engineer Waterways Experiment Station, Corps of Engineers, Vicksburg, Mississippi, January 1964. URL <https://digital.library.unt.edu/ark:/67531/metadc303901/>.

- [4] K. Gao, H. Wei, W. Xu, Z. Meng, X. Sun, and L. Sun. Simulation and analysis of propulsive performance for screw propulsion inspection robot. *Powder Technology*, 434, 2024. doi: 10.1016/j.powtec.2023.119300.
- [5] D. Osiński and K. Szykiedans. Small remotely operated screw-propelled vehicle. In R. Szewczyk, C. Zieliński, and M. Kaliczyńska, editors, *Progress in Automation, Robotics and Measuring Techniques*, volume 351 of *Advances in Intelligent Systems and Computing*, pages 191–200. Springer International Publishing, Cham, 2015. doi: 10.1007/978-3-319-15847-1_19.
- [6] S. J. Knight, E. S. Rush, and B. G. Stinson. Trafficability tests with the marsh screw amphibian. *Journal of Terramechanics*, 2(4):31–50, 1965. doi: 10.1016/0022-4898(65)90130-8.
- [7] M. J. Neumeyer and B. D. Jones. The marsh screw amphibian. *Journal of Terramechanics*, 2(4):83–88, 1965. doi: 10.1016/0022-4898(65)90133-3.
- [8] A. Thoesen, S. Ramirez, and H. Marvi. Screw-powered propulsion in granular media: An experimental and computational study. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, Brisbane, Australia, 2018.
- [9] A. Thoesen, T. McBryan, D. Mick, M. Green, J. Martia, and H. Marvi. Granular scaling laws for helically driven dynamics. *Physical Review E*, 102:032902, 2020. doi: 10.1103/PhysRevE.102.032902.
- [10] D. Chen, Z. Samuels, L. Peiros, S. Mukherjee, and M. C. Yip. Characterization and evaluation of screw-based locomotion across aquatic, granular, and transitional media. arXiv preprint, 2025. arXiv:2511.11958.
- [11] K. Sagara, D. Fujiwara, and K. Iizuka. Development and evaluation of a two-dimensional extension/contraction-driven rover for sideslip suppression during slope traversal. *Aerospace*, 12:699, 2025. doi: 10.3390/aerospace12080699.
- [12] M. Green, T. McBryan, D. Mick, D. Nelson, and H. Marvi. Regolith excavation performance of a screw-propelled vehicle. *Advanced Intelligent Systems*, 2021. doi: 10.1002/aisy.202100125.
- [13] F. Richter, P. V. Gavrilov, H. M. Lam, A. Degani, and M. C. Yip. ARCSnake: Reconfigurable snakelike robot with Archimedean screw propulsion for multidomain mobility. *IEEE Transactions on Robotics*, 38(2):797–809, 2022. doi: 10.1109/TRO.2021.3104968.
- [14] G. Genta and A. Genta. Preliminary assessment of a small robotic rover for Titan exploration. *Acta Astronautica*, 68(5–6):556–566, 2011. doi: 10.1016/j.actaastro.2010.02.016.
- [15] L. Matthies. NIAC phase 1 final study report on Titan aerial daughtercraft. Technical Report HQ-E-DAA-TN39363, NASA / Jet Propulsion Laboratory, 2017. URL https://www.nasa.gov/wp-content/uploads/2017/07/niac_2014_phase1_matthies_titanaerial_daughtercraft_tagged.pdf.
- [16] N. Deitrich, K. M. Baldonado, A. Khan, J. Cook, L. Rizzo, K. Schroeder, J. Bayandor, and J. Samareh. A rideshare tensegrity rover concept to explore Titan’s lands and oceans. Technical Report NASA/TM–20220014544, NASA Langley Research Center, October 2022. URL <https://ntrs.nasa.gov/api/citations/20220014544/downloads/NASA-TM-20220014544.pdf>.
- [17] K. R. Atkinson, J. C. Zarnecki, M. C. Towner, T. J. Ringrose, A. Hagermann, A. J. Ball, M. R. Leese, G. Kargl, M. D. Paton, R. D. Lorenz, and S. F. Green. Penetrometry of granular and moist planetary surface materials: Application to the Huygens landing site on Titan. *Icarus*, 210(2):843–851, 2010. doi: 10.1016/j.icarus.2010.07.019.
- [18] C. A. Nixon et al. Titan’s cold case files — outstanding questions after Cassini-Huygens. *Planetary and Space Science*, 155:50–72, 2018. doi: 10.1016/j.pss.2018.02.009.
- [19] R. D. Lorenz. Titan Bumblebee: A 1 kg lander-launched UAV concept. *Journal of the British Interplanetary Society*, 61:118–124, 2008. URL <https://bis-space.com/shop/product/titan-bumblebee-a-1-kg-lander-launched-uav-concept/>.
- [20] S. Rodriguez et al. Science goals and new mission concepts for future exploration of Titan’s atmosphere, geology and habitability: titan POLar scout/orbitEr and in situ lake lander and DrONe explorer (POSEIDON). *Experimental Astronomy*, 54(2–3):911–973, 2022. doi: 10.1007/s10686-021-09815-8.
- [21] Z. Q. Li and L. K. Bingham. NASA white paper — terramechanics for LTV modeling and simulation. White paper, NASA Johnson Space Center, 2022. URL https://ntrs.nasa.gov/api/citations/20220010732/downloads/Terramechanics_white_paper.pdf. NASA Technical Reports Server document 20220010732.
- [22] N. Patel. *An investigation into optimal mobility systems for planetary rovers*. PhD thesis, Surrey Space Centre, University of Surrey, 2005.
- [23] S. Wakabayashi, H. Sato, and S.-I. Nishida. Design and mobility evaluation of tracked lunar vehicle. *Journal of Terramechanics*, 46:105–114, 2009.